J-space Phase 4 — development synthesis

OLMo training trajectory and coordinate-frame validity

State through OLMo-3 32B base and the seed-paired 3.0/3.1 Think lens comparisons · 2026-07-31

Scope. This is a living **development only** report over known Phase 3 banks. It localizes trajectory patterns and audits lens-frame dependence; it cannot establish a binary lineage claim. No Phase 4 confirmatory or replication cell is licensed before PI sign-off and a tagged preregistration freeze.

1 Compute and provenance boundary

Every model producer hard-failed unless CUDA was visible in the same process, executed a finite FP16 CUDA matrix multiply before model load, and recorded the device in its result envelope. Jobs ran on an NVIDIA RTX PRO 6000 Blackwell Server Edition; the final grid allocated about 81.1 GB VRAM. Model loading, generation, interventions, and scoring never used CPU. CPU was limited to tests, hashes, statistics, figures, and document compilation.

Phase 3 enters through the immutable `p4-import-phase3-release-v1`. Native evidence is event-sourced; registered artifacts are never overwritten.

2 Prospective capability gate

OLMo-3 32B Think produced 972 G5 rows, 324 facts, and 72 families under prospective prefix-disjoint accepted-alias scoring.

view	overall	Bank F	Bank S	direct	composed	bridge supplied
capability	.6430	.4935	.8972	.6574	.4846	.7870

The intervention cohort was fixed before outcomes by requiring both direct and composed generation capability within a fact: 41/204 Bank-F facts and 85/120 Bank-S facts, or 126 facts / 252 items.

Evidence: `p4-g5-bank-olmo3-think-dev-v2`

The first attempt stopped after accent folding made `Río` and `Rio` ambiguous. The repaired selector prefers exact spelling before a unique normalized fallback; the partial attempt remains preserved.

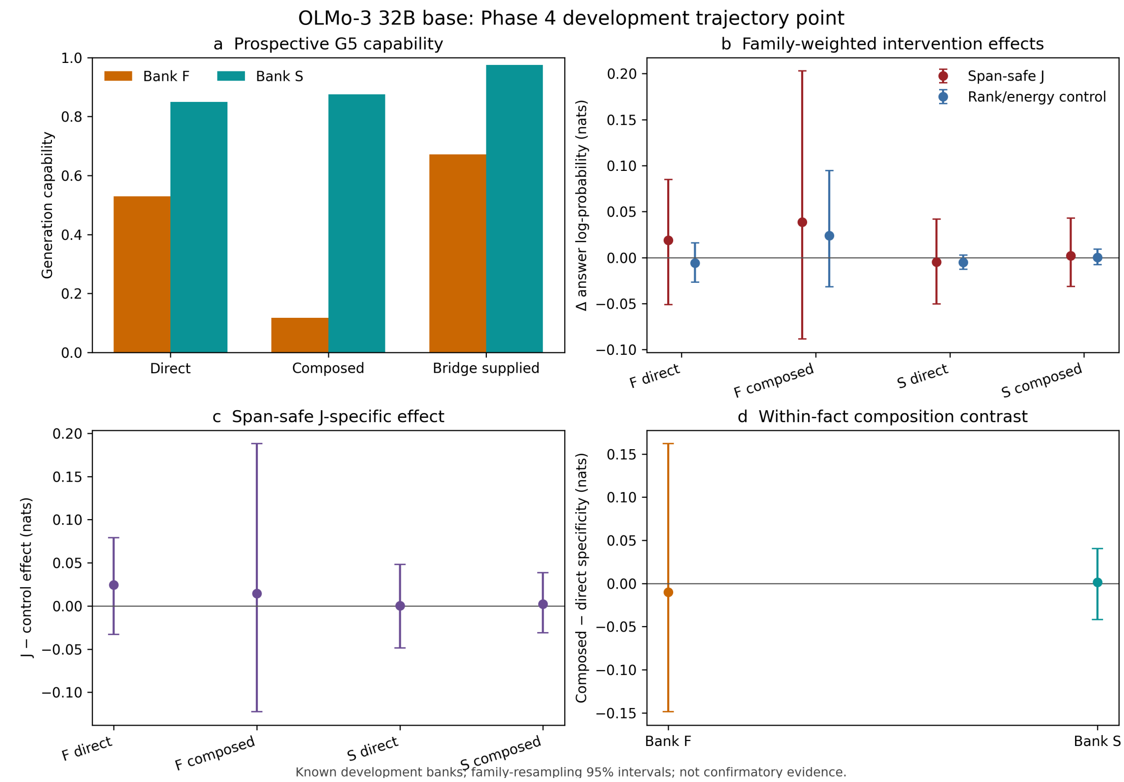
3 OLMo-3 32B base point

The exact pinned base revision `c2b61dae89a1ad10e4ad5653d0e46b590902607b` produced 972 G5 rows. Overall capability was .6101 (Bank F .4395; Bank S .9000). Requiring both direct and composed capability fixed 23 Bank-F and 88 Bank-S facts, or 111 facts / 222 items.

Evidence: `p4-g5-bank-olmo3-base-dev-v1`

The seven-condition grid had maximum baseline replay drift 6.71×10^{-7} nats, zero span-safe selected/protected overlap, 100% matched-rank agreement, and maximum energy relative error 0.000260.

Evidence: `p4-lineage-grid-olmo3-base-dev-v1`



cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	+0.0245	[−0.0330, +0.0792]
Bank F composed	+0.0145	[−0.1226, +0.1881]
Bank S direct	+0.0005	[−0.0487, +0.0481]
Bank S composed	+0.0021	[−0.0309, +0.0385]

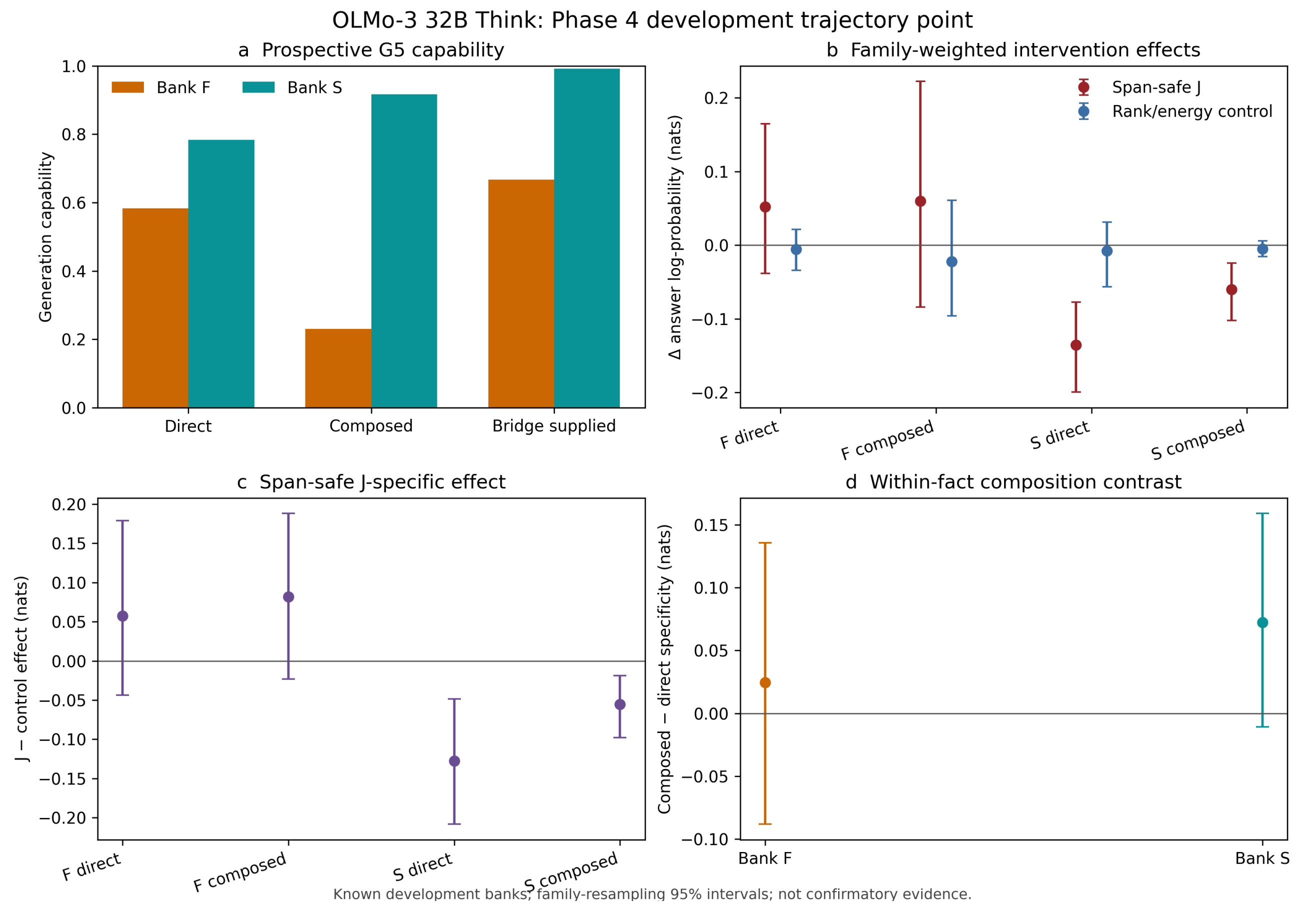
All four primary effects are near zero. Bank-S composed-minus-direct is +0.0016 [−0.0417, +0.0403]. The experiment is nevertheless sensitive: overall label-protected J-specific is +0.1231 [+0.0667, +0.1802], while mechanics-random and logit-protected effects are −0.1048 and −0.2023, with intervals below zero.

Evidence: `p4-lineage-analysis-olmo3-base-dev-v1`

4 Think own-lens development point

The seven-condition grid includes baseline, span-safe J, exact rank/energy control, label-protected J, protected-energy control, mechanics random, and logit-space label protection. Baseline replay drift is at most 7.15×10^{-7} nats. Rank agreement is 100%, energy relative error is at most 0.000230, and span-safe selected/protected overlap is zero.

Evidence: `p4-lineage-grid-olmo3-think-dev-v1`



cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	+0.0574	[−0.0442, +0.1785]
Bank F composed	+0.0818	[−0.0223, +0.1878]
Bank S direct	−0.1277	[−0.2072, −0.0494]
Bank S composed	−0.0553	[−0.0982, −0.0187]

Bank-S composed-minus-direct specificity is +0.0724 [−0.0109, +0.1584]. It is compatible with the unusual 3.1 Think tendency beginning by 3.0, but is not resolved and must await the controlled Bank-W redundancy/derivation study.

Evidence: `p4-lineage-analysis-olmo3-think-dev-v1`

5 Common-base-lens repair audit

The common coordinate is the frozen OLMo base lens (`92f32e...b696`). V1 exposed rank-limited protected-energy sites that omitted one requested component without being marked clamped. V2 repaired the metadata but revealed that the evidence ID itself changed all matched/random seeds.

V3 freezes the scientific-seed namespace to v1. All seven aggregate outcome columns, every per-alias score, and every condition order match v1 exactly (maximum difference 0). The fix reclassifies 296 summaries and reduces the erroneous maximum unclamped energy error from 0.511283 to 0.000239 while retaining 100% rank agreement. V3 supersedes both withdrawn diagnostic attempts and remains valid standalone evidence.

Evidence: `p4-lineage-grid-olmo3-think-common-base-lens-dev-v3`

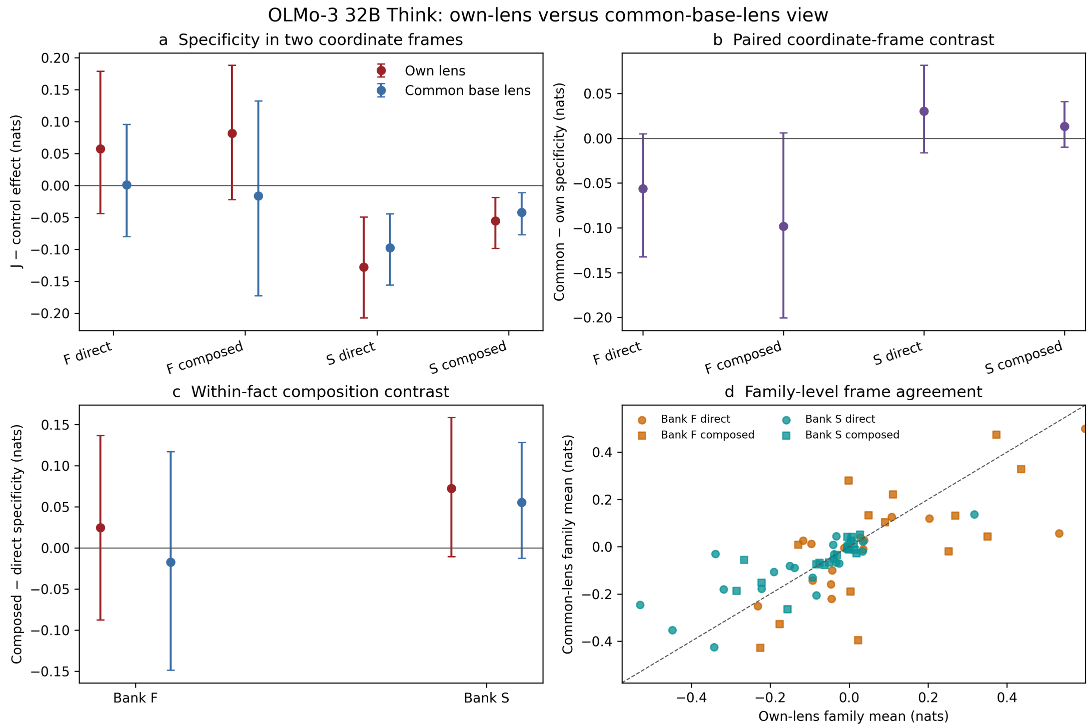
V4 reruns the common lens under the own-grid seed namespace. Its 252 rows form 126 exact direct/composed pairs; condition order matches the own grid item-for-item, between-frame baseline drift is zero, matched rank agreement is 100%, maximum energy error is 0.000205, and span-safe overlap is zero. Deterministic J arms match v3 exactly.

Evidence: `p4-lineage-grid-olmo3-think-common-base-lens-dev-v4`

6 Own/common lens comparison: corrected

The v1 paired analysis remains withdrawn because its frames used different matched-control RNG streams. V2 shares the exact namespace and condition order. Its full 100,000-draw payload and all bootstrap hashes reproduce exactly. The hardened producer refuses unequal namespaces.

Evidence: `p4-lens-frame-analysis-olmo3-think-dev-v2`



Same 126 paired facts; known development banks; family-resampling 95% intervals; not confirmatory evidence.

cell	own lens	common lens	common − own (95% interval)
F direct	+0.0574	+0.0010	−0.0563 [−0.1324, +0.0049]
F composed	+0.0818	−0.0164	−0.0982 [−0.2005, +0.0058]
S direct	−0.1277	−0.0974	+0.0303 [−0.0165, +0.0813]
S composed	−0.0553	−0.0419	+0.0134 [−0.0100, +0.0408]

No frame-delta interval excludes zero. Bank-S direct and composed effects have intervals below zero in each tested frame; Bank-F effects remain imprecise. Item and family frame correlations are .7557 and .7254; mean absolute item difference is .1025 nats. The RNG correction leaves J deltas unchanged, moves Bank-F specificity deltas 0.0413 and 0.0492 nats toward zero through the controls, and raises family correlation from .4946 to .7254.

7 OLMo-3.1 Think capability and own frame

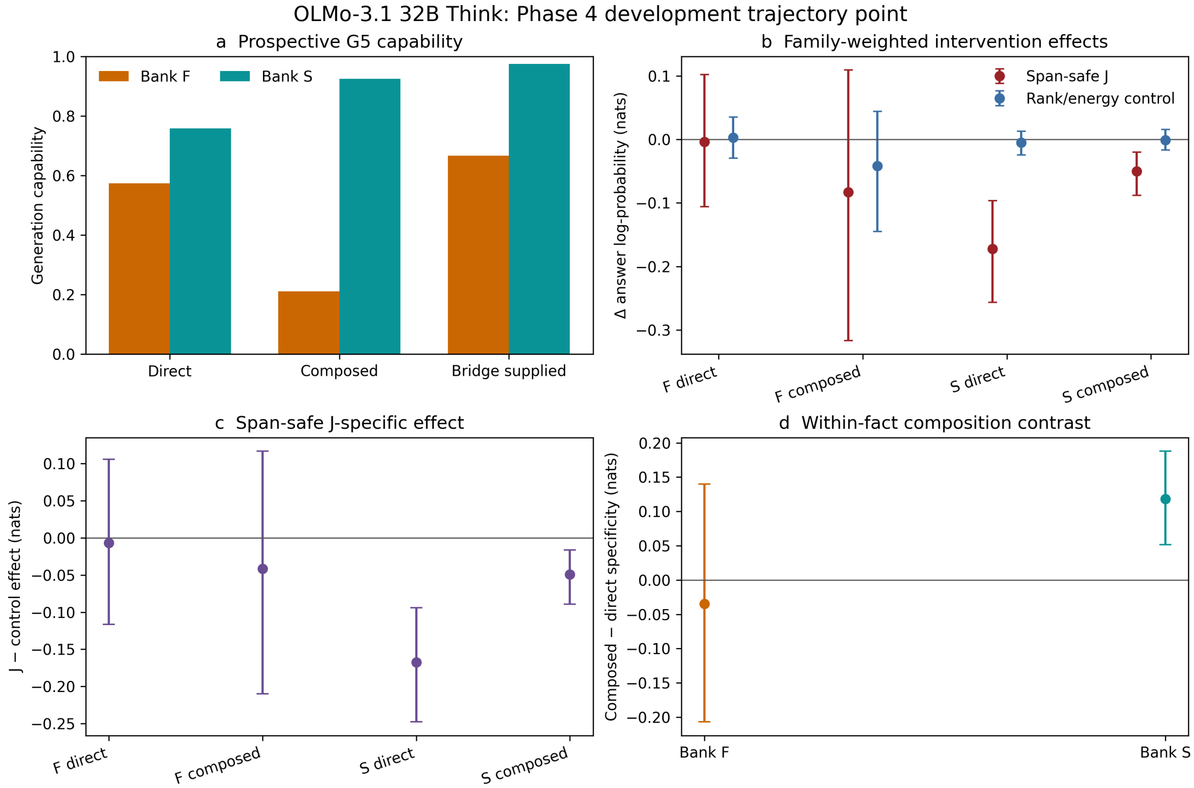
The exact pinned 3.1 Think revision `832c3f54...7840544` produced 972 G5 rows. Boundary-safe prefix capability was .6327 overall (Bank F .4837; Bank S .8861), with direct .6420, composed .4753, and bridge-supplied .7809. Requiring both direct and composed capability fixed 38 Bank-F and 84 Bank-S facts, or 122 facts / 244 items.

Evidence: `p4-g5-bank-olmo31-think-dev-v1`

This rate is below Phase 3's historical .7346 because that cohort allowed an answer anywhere in the continuation. All 972 generations are byte-identical across phases, and every Phase 4 flag equals the later Phase 3 boundary-safe prefix audit.

The own-lens grid has maximum G5 replay drift 8.64×10^{-7} nats, zero span-safe overlap, 100% rank agreement over 564 matched summaries, and maximum energy error 0.000222.

Evidence: `p4-lineage-grid-olmo31-think-dev-v1`



Known development banks; family-resampling 95% intervals; not confirmatory evidence.

cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	-0.0067	$[-0.1162, +0.1057]$
Bank F composed	-0.0416	$[-0.2099, +0.1167]$
Bank S direct	-0.1674	$[-0.2477, -0.0942]$
Bank S composed	-0.0491	$[-0.0891, -0.0165]$

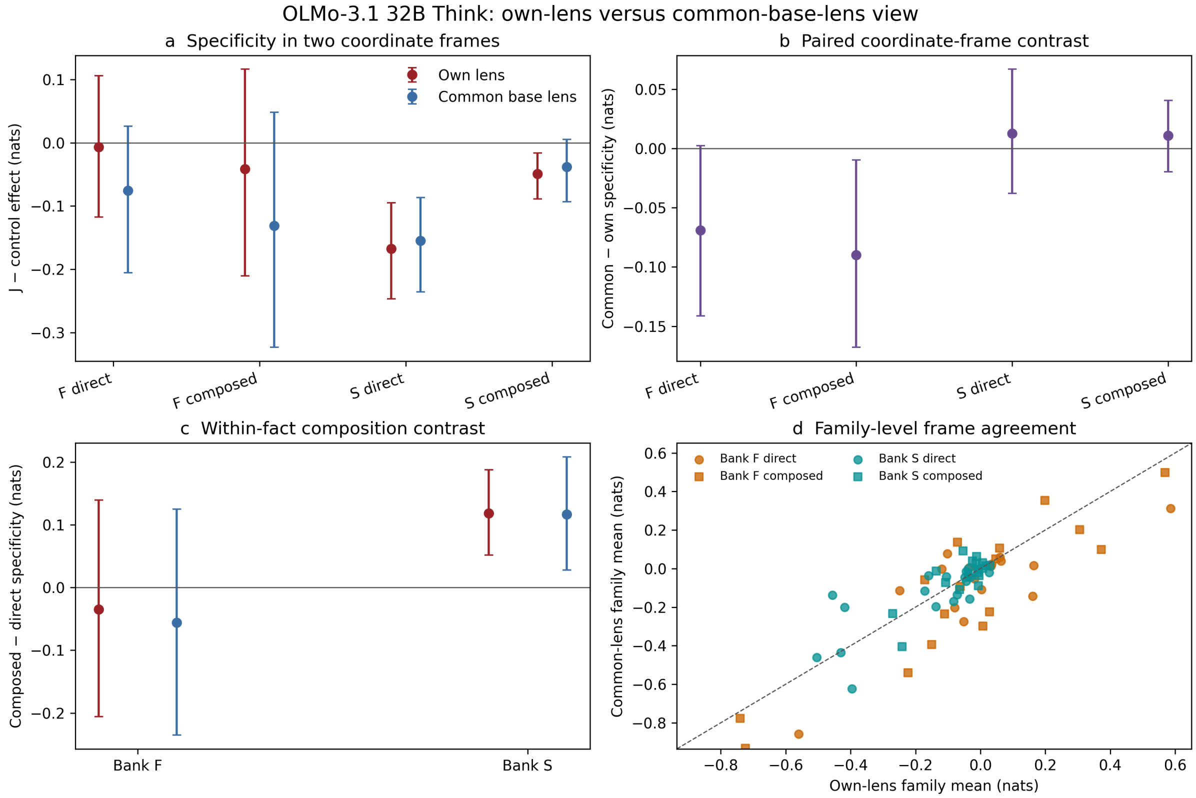
Bank-S composed-minus-direct is $+0.1183\,[+0.0514, +0.1879]$ (descriptive exact sign-flip $p = .0031$). Analysis v1 has an identical numerical payload but is withdrawn for a crowded figure footer; v2 fixes the reusable layout.

Evidence: `p4-lineage-analysis-olmo31-think-dev-v2`

8 OLMo-3.1 Think paired coordinate comparison

The common grid uses the frozen base lens under the exact own-grid seed namespace. All 244 item IDs and condition orders pair; between-frame baseline drift is zero. Its maximum matched energy error is 0.000256, with zero span-safe overlap and 100% rank agreement. Mechanics-random and logit-protected outcomes are bit-identical between frames.

Evidence: `p4-lineage-grid-olmo31-think-common-base-lens-dev-v1`



Same 122 paired facts; known development banks; family-resampling 95% intervals; not confirmatory evidence.

cell	own lens	common lens	common − own (95% interval)
F direct	−0.0067	−0.0757	−0.0690 [−0.1413, +0.0025]
F composed	−0.0416	−0.1314	−0.0898 [−0.1677, −0.0096]
S direct	−0.1674	−0.1547	+0.0127 [−0.0378, +0.0669]
S composed	−0.0491	−0.0381	+0.0110 [−0.0196, +0.0406]

Only the Bank-F composed frame-delta interval excludes zero: evidence of coordinate sensitivity for that known-bank cell, although the effect itself remains imprecise in each frame. Bank-S direct is negative in both frames. The Bank-S composition contrast is positive in both frames: own +0.1183 [+0.0517, +0.1876], common +0.1166 [+0.0279, +0.2080]; paired delta −0.0017 [−0.0551, +0.0504]. Item/family correlations are .7679/.8386 and mean absolute item shift is .1133 nats. All 42 bootstrap and 6 exact sign-flip distributions reproduce exactly.

Evidence: `p4-lens-frame-analysis-olmo31-think-dev-v1`

9 Interpretation and next boundary

1. The base checkpoint is near zero in all four primary cells. Bank-S direct and composed specificity is negative by 3.0 Think and is negative with intervals below zero in both seed-paired frames. The paired frame deltas include zero: development evidence of two-frame robustness, not a confirmatory invariance claim.
2. Because capability cohorts are checkpoint-specific, this localizes a base-to-Think trajectory change but is not a paired causal estimate.
3. At 3.1 Think, Bank-S direct remains negative in both frames. Bank-S composed is negative in the own frame but imprecise in the common frame. The positive composition contrast is precise in both frames and stable under the lens change.
4. Bank-F cell effects remain imprecise, but the 3.1 composed common-minus-own interval is below zero. Coordinate choice materially affects at least one known-bank trajectory cell.
5. The 3.1 composition result is a development finding, not a mechanism; controlled Bank-W load/redundancy and internal-derivation axes remain necessary.
6. None of these known-bank estimates licenses "lineage present" or "lineage absent."

The base and corrected 3.0/3.1 Think paired points are complete and banked. Next, run exact 3.1 Instruct through the prospective G5 gate and seed-paired own/common seven-condition grids. Then synthesize the full base → 3.0 Think → 3.1 Think/Instruct trajectory. If the two coordinate views disagree, prioritize the fit-size/corpus study before stronger causal interpretation.